CURRICULUM VITAE

Gerald S. Macaraig

Computer Engineering Graduate | Python & Rust Developer | AI Systems & Automation

Cabuyao City, Laguna, Philippines | macaraiggerald97@gmail.com | 0998 169 6047

PROFESSIONAL SUMMARY

Computer Engineering graduate with experience in corporate IT support, semiconductor manufacturing operations, and software projects involving Python, Rust, AI-assisted error analysis, retrieval pipelines, and computer vision. Worked on Gemma LLM workflows, BM25 ranking, vector search tuning, and developer tools for crash analysis and troubleshooting. Combines a practical engineering mindset with experience in network configuration, hardware and software troubleshooting, inventory operations, and circuit-related systems. Interested in software development, automation, AI applications, and IT systems.

TECHNICAL SKILLS

Programming & Scripting: Python, Rust, C/C++

AI & Retrieval: Gemma LLM workflows, AI-assisted crash analysis, BM25 ranking, vector search, Retrieval-Augmented Generation (RAG)

Computer Vision & OCR: YOLOv11n, OpenCV, Tesseract OCR, image processing

Hardware & Engineering: Raspberry Pi, Arduino, microcontrollers, sensors, PCB design, soldering, circuit assembly, robotics

Tools & Applications: Git, GitHub, VS Code, Microsoft Excel, KiCad, AutoCAD, Proteus, Fusion 360

IT & Operations: Network configuration, IP address assignment, hardware/software troubleshooting, OS installation and reformatting, inventory documentation

PROFESSIONAL EXPERIENCE

Rust Programmer / AI Systems Developer **2026**

- Developed AI-assisted error-handling solutions for mobile applications.
- Integrated Gemma LLM workflows to support automated crash analysis.
- Implemented BM25 retrieval pipelines and tuned vector search systems to improve error matching.
- Assisted in identifying, classifying, and resolving application crashes through AI-assisted analysis.
- Built developer productivity tools to accelerate troubleshooting and support software reliability.

IT Support Intern | Vmpowered Corp. **2025**

- Configured networks and managed IP address assignments; assisted with connectivity, access, and network performance issues.
- Troubleshot hardware and software issues, documented recurring network problems, and supported OS installation and reformatting.

Customer Sales Representative | Vmpowered Corp. **2022**

- Supported customer accounts across healthcare, mortgage, loan, and car insurance services.
- Handled inquiries, policy updates, and claims processes; documented interactions and worked toward account service and sales targets.

Material Handler | Nexperia Inc. **2021**

- Supported material flow through molding; managed raw materials, components, and finished goods for semiconductor production.
- Coordinated material supply with production teams and maintained transfer and stock-level documentation.

SELECTED WORK & EDUCATION

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SELECTED PROJECTS

Smart Schematic Diagram Analyzer for Electrical Circuits Using Computer Vision and a RAG-Based Feedback System

Academic Capstone | Python, YOLOv11n, OpenCV, Tesseract OCR, BM25, RAG, Raspberry Pi

Developed a prototype concept for analyzing hand-drawn and printed electrical schematics, detecting components, verifying circuit structure, and providing retrieval-based corrective guidance. Integrated camera-based capture and Raspberry Pi hardware with computer vision and a local BM25 knowledge-retrieval workflow.

AI-Assisted Crash Analysis and Debugging Automation

AI Systems Development | Rust, Gemma LLM, BM25, Vector Search

Worked on AI-assisted mobile application error handling, automated crash analysis workflows, retrieval-based error matching, and developer productivity tools for faster troubleshooting.

Asset Monitoring System

Inventory Management | Excel Integration

Worked on an asset-monitoring solution focused on organizing inventory records and supporting asset tracking through structured data and Excel integration.

EDUCATION

Bachelor of Science in Computer Engineering 2022–2026
University of Cabuyao (Pamantasan ng Cabuyao)

Senior High School, ICT (Information and Communications Technology) 2018–2020
Citi Global Colleges

CERTIFICATIONS

- Cisco – AI Fundamentals with IBM SkillsBuild
- University of the Philippines Open University (UPOU) – Basics of Resilience